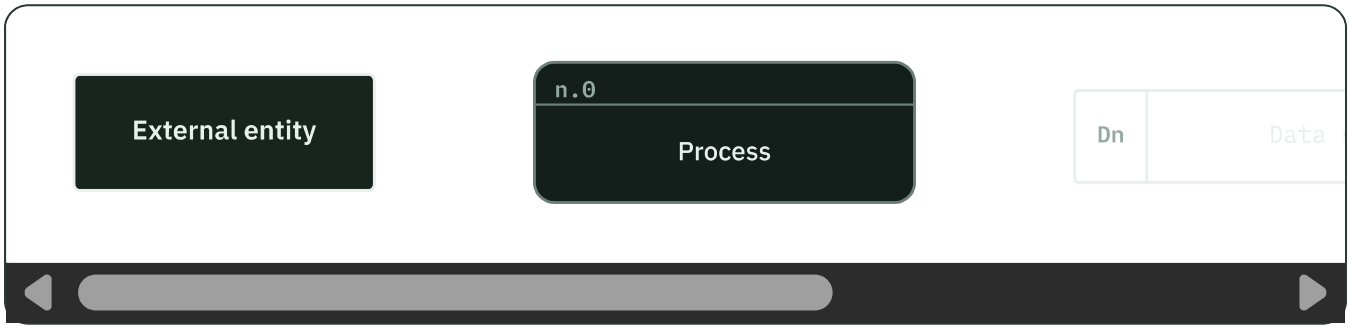
8 processes · 8 data stores

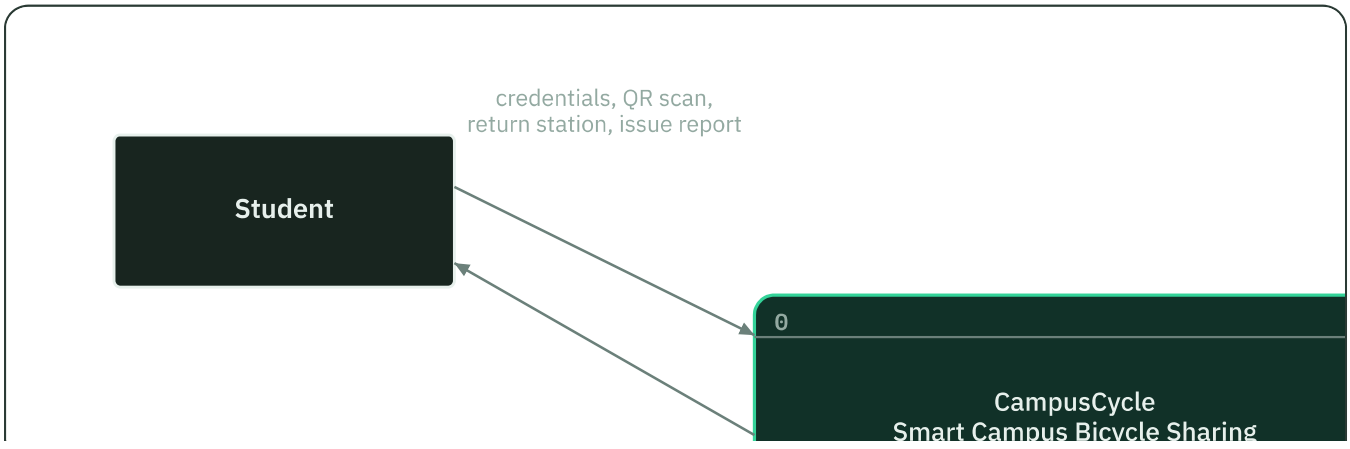


Notation. Gane–Sarson symbols. A dashed outline marks something that does not exist yet in the built system.

Diagrams scroll sideways on a narrow screen.

Context diagram

At the boundary the system has two kinds of user and one honest gap. Everything a student or an administrator sends arrives through the REST API with a signed token; the identity on that token, never an id in the request body, decides what comes back.



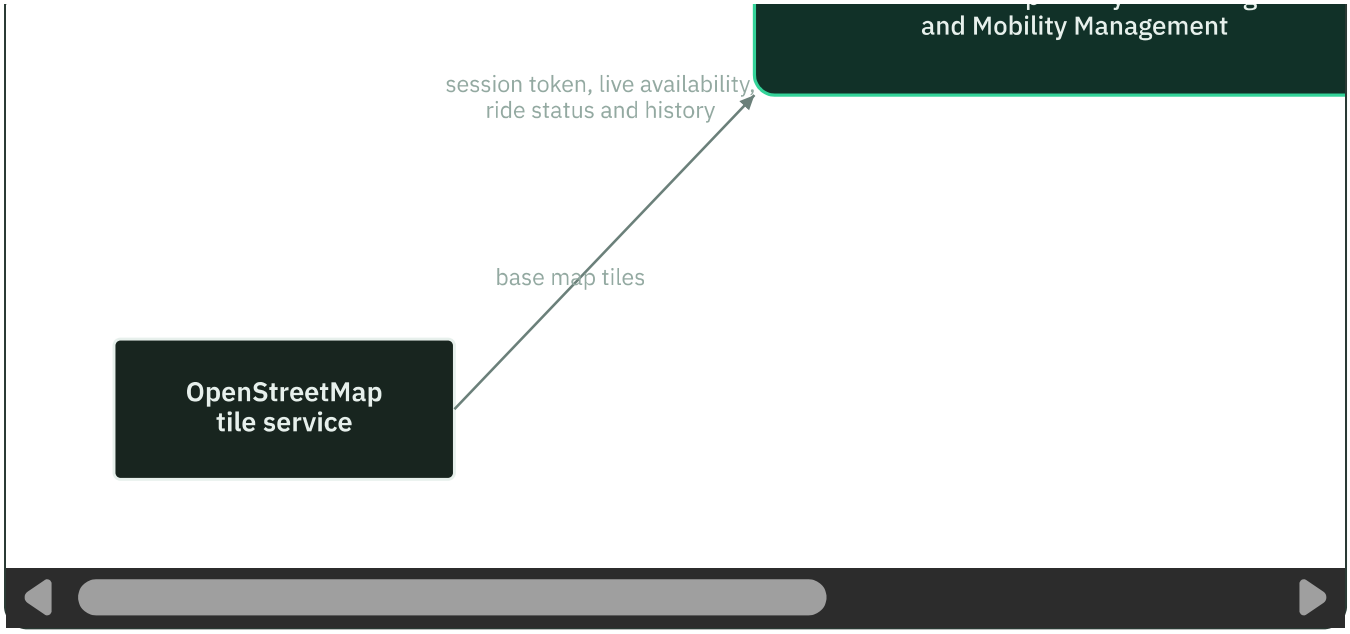
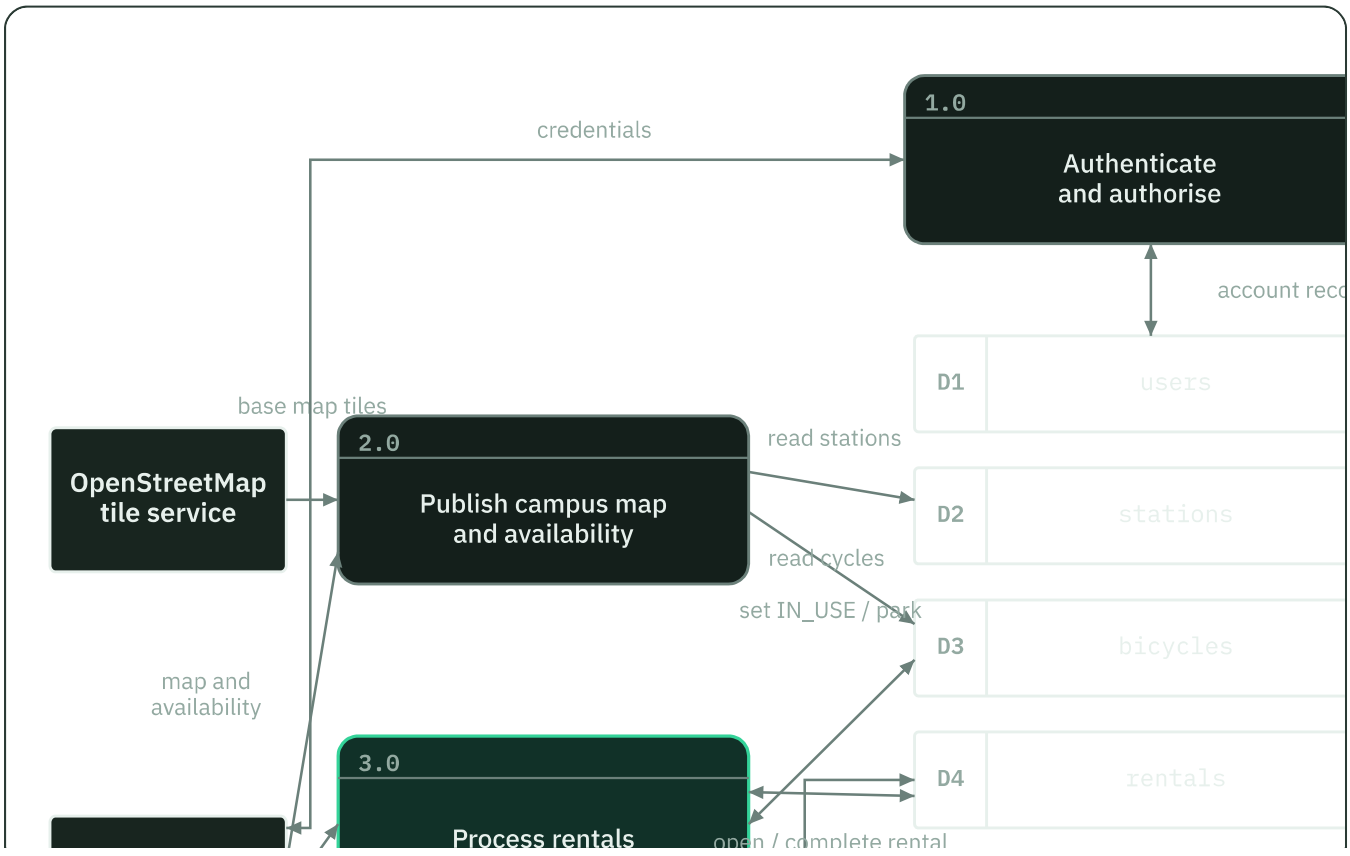


Figure 1 · Context diagram (Level 0). Who the system exchanges data with. The dashed feed is deliberate: no GPS hardware exists on the bicycles, so every position the system holds was submitted by an administrator and stored as `SIMULATED`.

Level 1

Eight processes, each owning the tables it writes. Reads cross freely — analytics reads almost everything — but a write to a store belongs to one process, which is what keeps the state machine for a cycle in one place.



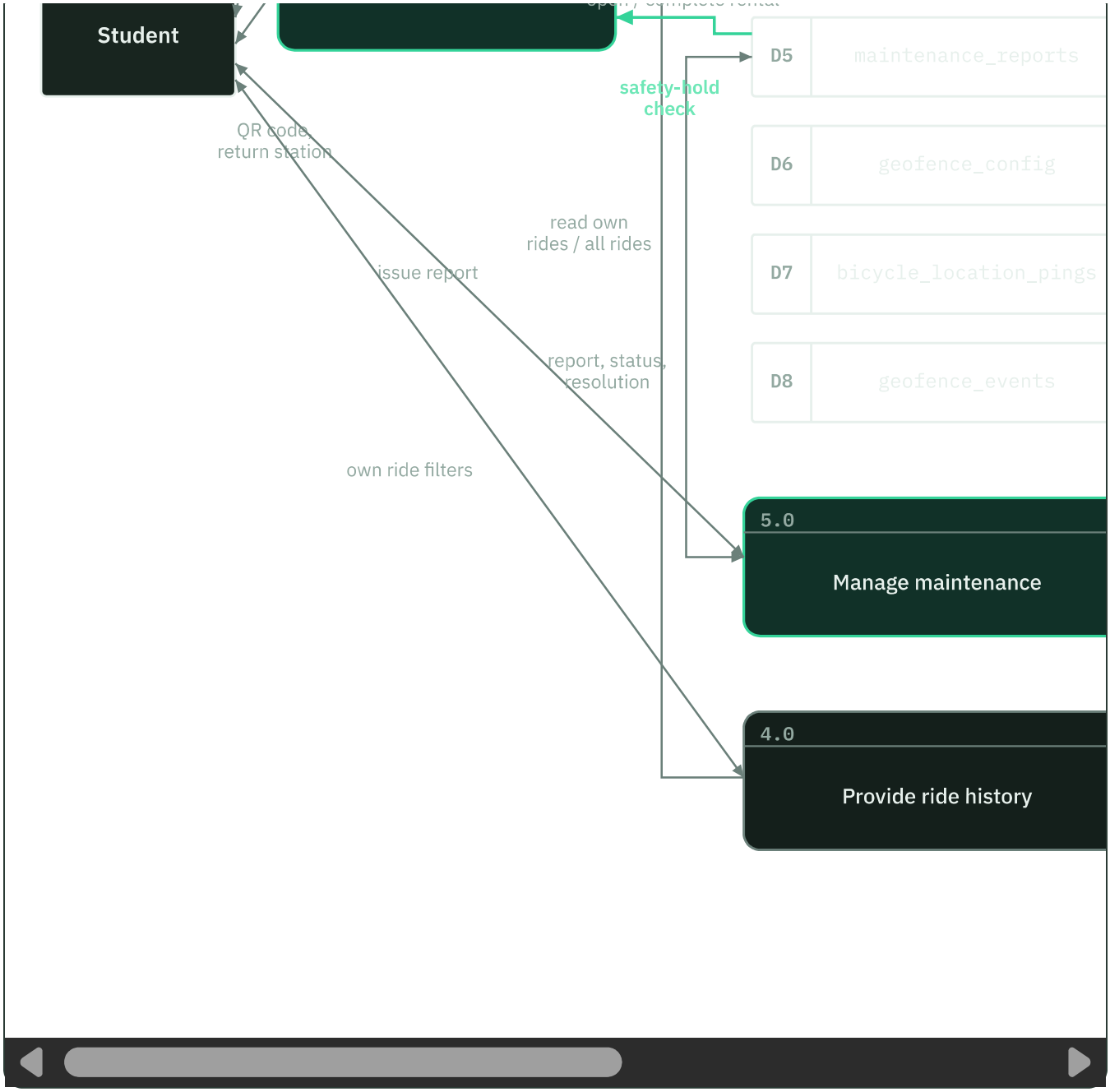
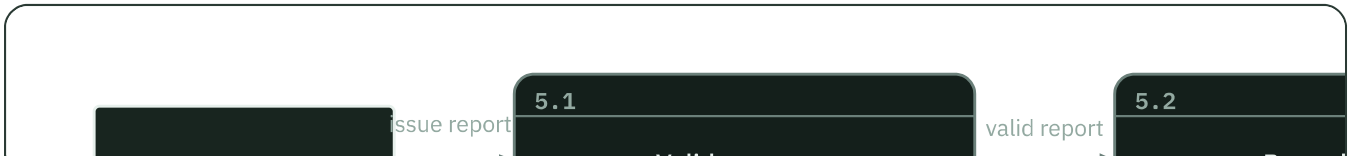


Figure 2 · Level 1. Eight processes around the eight tables they use. Students enter on the left, administrators on the right, and the three processes both roles reach sit above and below the spine. The green flows are the ones the fleet's safety rests on: 5.0 writes a hold onto a cycle and 3.0 must read it before opening a rental.

Level 2 · maintenance and the safety hold

This is the part of the system worth exploding: a reported fault has to reach the rental path, or a cycle a student called unsafe stays rentable. Severity is what connects them.



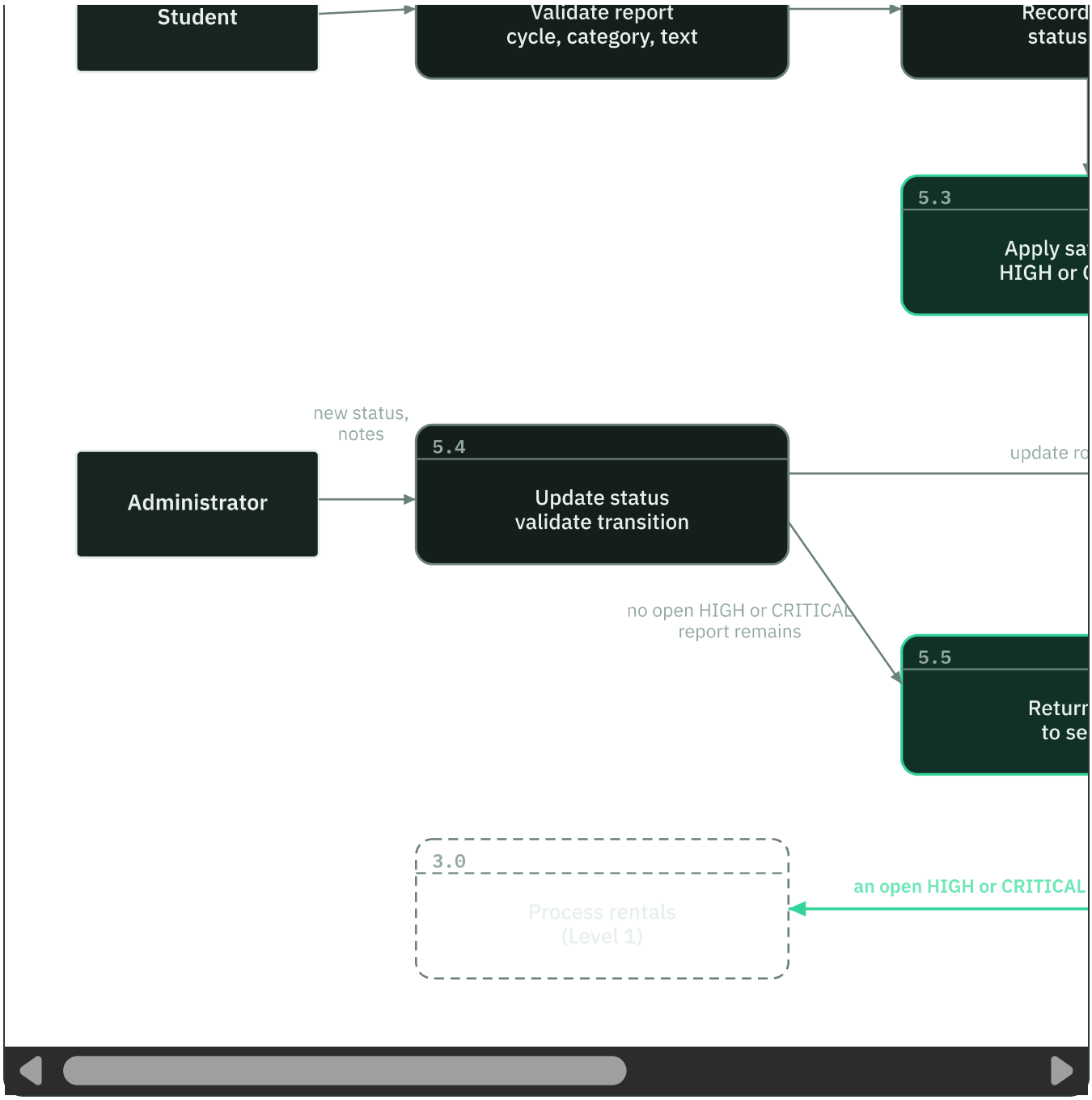


Figure 3 · Level 2 — process 5.0, maintenance and the safety hold. The green path is the invariant the system is built around: a cycle with an unresolved HIGH or CRITICAL report cannot be rented and cannot be set back to AVAILABLE by any route. Resolving the report does not return the cycle to service on its own — 5.5 is a separate, explicit admin action.

Level 2 · geofencing

The second explosion covers the feed that does not exist yet. Because positions are submitted rather than measured, the source is validated first and recorded on every row, and de-duplication is what keeps one wandering cycle from filling the alert list.

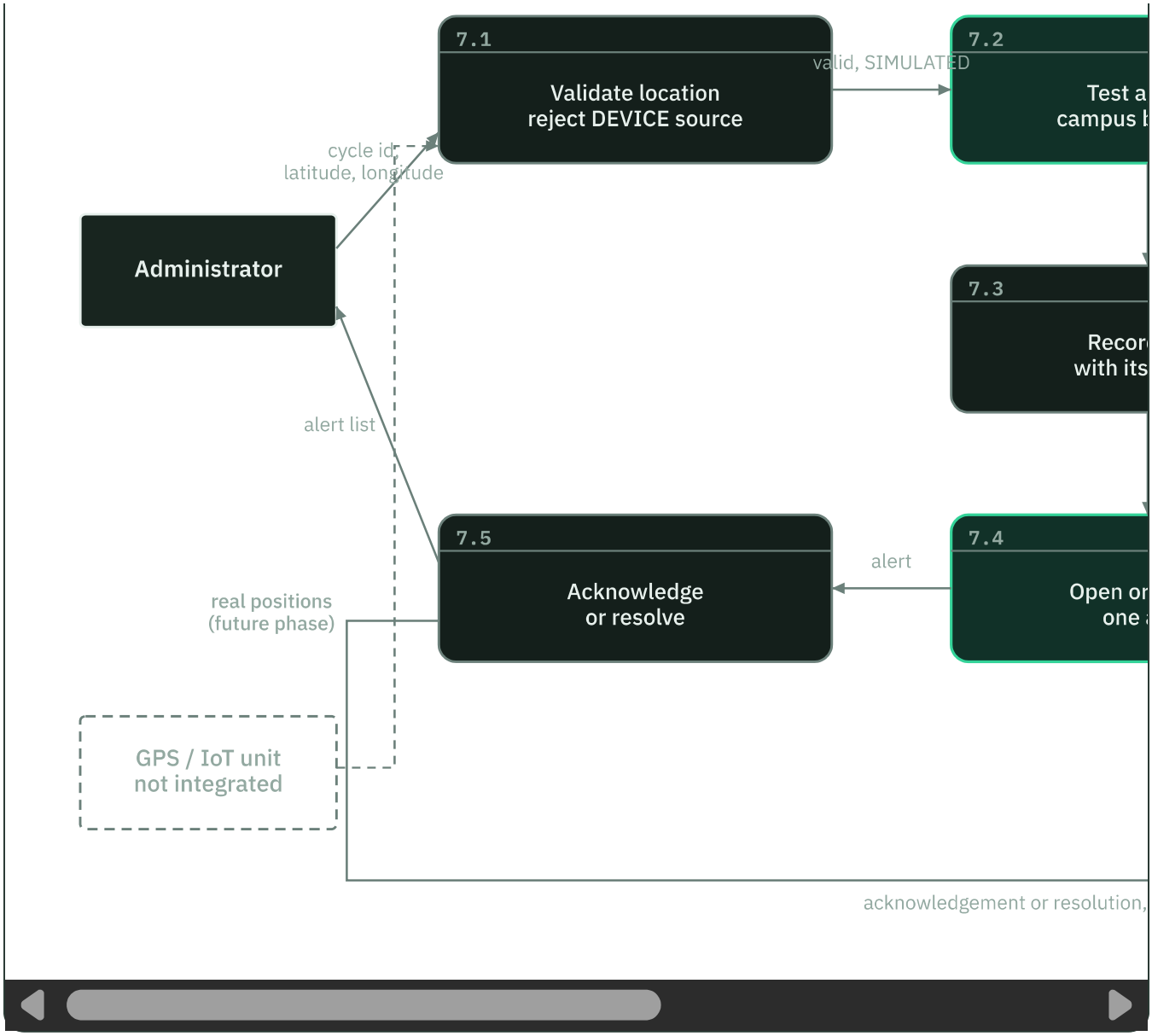


Figure 4 · Level 2 — process 7.0, geofencing. 7.1 refuses anything not marked SIMULATED, so a simulated position can never be filed as a real one. 7.4 is where repeated breaches stop becoming noise: while a cycle already has an unresolved alert, a further out-of-bounds ping increments that alert instead of opening another.

Data stores

Each store is one PostgreSQL table. The owning process is the only one that writes to it.

STORE	TABLE	WHAT IT HOLDS	WRITTEN BY
D1	users	Student and admin accounts, bcrypt hash, role.	1.0
D2	stations	Name, location, capacity, status, latitude and	6.0

		longitude.	
D3	bicycles	Cycle code, QR code, current station, status.	6.0
D4	rentals	One row per ride: rider, cycle, start and end station, timestamps, status.	3.0
D5	maintenance_reports	Reported issue, severity, status, reporter, resolution and who closed it.	5.0
D6	geofence_config	The campus boundary polygon. One row active at a time.	7.0
D7	bicycle_location_pings	Every submitted position, with the source that supplied it.	7.0
D8	geofence_events	Boundary violations. At most one unresolved row per cycle.	7.0

Processes

How each process on the Level 1 diagram maps onto the code, for anyone reading the two side by side.

NO.	PROCESS	READS	WRITES	IMPLEMENTED IN
1.0	Authenticate and authorise	D1	D1	authService, middleware/auth.js
2.0	Publish campus map and availability	D2, D3, D6	—	stationService.mapData
3.0	Process rentals	D3, D5	D3, D4	rentalService
4.0	Provide ride history	D4	—	rentalService.getHistoryForUser, listPaged
5.0	Manage maintenance	D3, D5	D3, D5	maintenanceService
6.0	Manage stations and fleet	D2, D3	D2, D3	stationService, bicycleService
7.0	Monitor geofence	D6, D8	D7, D8	geofenceService

8.0	Generate analytics	D2, D3, D4, D5	—	analyticsService

Two flows carry the system's guarantees

Safety hold. 3.0 reads D5 before it opens a rental, and 5.0 is the only process that clears the hold. A cycle with an unresolved HIGH or CRITICAL report cannot be rented, cannot be parked as AVAILABLE when it is returned, and cannot be forced back into service by an administrator.

Alert de-duplication. 7.4 updates an existing unresolved row rather than inserting a new one. A partial unique index on `geofence_events` enforces that in the database, so the rule holds even if two position updates arrive at once.